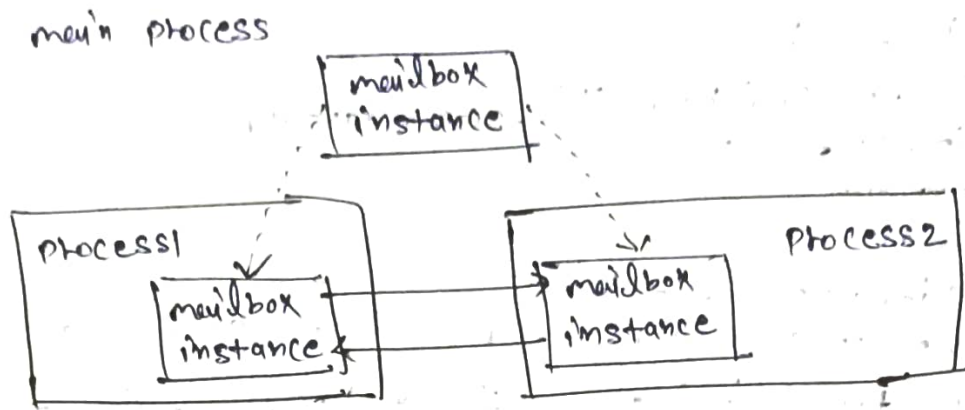


- in systemverilog a mailbox is a built in class used for inter process communication (IPC), while Semaphore only send empty synchronization signal (keys), mailboxes allow parallel threads to actually pass data packets back and forth.
- conceptually a mailbox works like a FIFO queue. one thread deposits a data packet into the mailbox and another thread retrieves it.



Bounded vs unbounded mailboxes:-

unbounded mailbox (new()): Has infinite capacity. the putting thread will never block BCZ the mailbox can always hold more data.

Bounded mailbox (new(bound)): Has a fixed size limit. If the mailbox fill up, any thread trying to put more data inside will block (wait) untill receiving thread removes an item to clear up space.

Core methods

method type

to send data (put)

to receive data (get/peek)

Blocking
(wait if full/empty)

put(item);

get(variable);
peek(variable);

Non-blocking
(Return 1 on success, 0 on Failure)

try-put(item);

try-get(variable);
try-peek(variable);

get(): retrieves an item and remove it from the mailbox queue.

peek(): copies the item but leaves it inside the mailbox.

num(): Return the current number of messages sitting inside the mailbox.